Ayushman Bharat Healthcare Analytics: Analysis of Hospital Accessibility and Services

Project-Based Learning (PBL) mini project for **Data Analytics using Python (CSL-0312)**, B.Tech. CSE (AIML), Semester III, School of Computing and Intelligent Systems, ITM University Gwalior. Academic session: July–Dec 2026.

Faculty Mentor: Dr. Sanjay Jain

Team

Name	Roll No.	Role
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About the project

Beneficiaries with an Ayushman Bharat (PM-JAY) card often find it hard to know which empanelled hospitals are available in their district, what type they are, and what specialties they offer. This project cleans and analyses the Madhya Pradesh empanelled hospital data (1,230 records, 51 districts) to show how hospitals are distributed and where coverage is comparatively low.

Dataset

- **File:** `Data_sets_of_data_analytics.xls` (1,230 records, 18 columns in the original file)
- **Source:** PM-JAY Hospital Search Portal, National Health Authority — https://hospitals.pmjay.gov.in/Search/empnlWorkFlow.htm
- **Cleaned file:** `ayushman_mp_hospitals_clean.csv` (1,230 records, 12 columns, including 4 derived columns)

Repository contents

File	Description
`ayushman_bharat_analysis.ipynb`	Jupyter Notebook with cleaning, EDA, visualizations and statistics
`ayushman_mp_hospitals_clean.csv`	Cleaned dataset
`Data_sets_of_data_analytics.xls`	Original dataset
`PBL_Report_Ayushman_Bharat_Healthcare_Analytics.docx`	PBL report
`fig1_top_districts.png` to `fig4_district_type.png`	Charts generated by the notebook

How to run

1. Install Python 3 and the libraries: `pip install pandas numpy matplotlib xlrd jupyter`
2. Keep `Data_sets_of_data_analytics.xls` in the same folder as the notebook.
3. Open `ayushman_bharat_analysis.ipynb` in Jupyter Notebook and choose **Run All**.

Tools

Python, Pandas, NumPy, Matplotlib, Jupyter Notebook, Microsoft Excel.

Key findings

- Hospitals are concentrated in a few districts: Bhopal has 256 (20.8%), and the top 3 districts hold 38.4% (Gini coefficient 0.497).
- Private (For Profit) hospitals are 49.7% overall and dominate Bhopal; public hospitals dominate smaller districts such as Rewa.
- 14 districts have 9 or fewer empanelled hospitals.
- 154 records (12.5%) are empanelled only under CGHS, not PM-JAY.

Counts are not adjusted for population, so they show availability and not hospitals per person.

AI tool usage

Claude (Anthropic) was used for drafting the report text and writing and running the analysis code. See Section 10 of the report.